

M1.4: Circular Motion

1. Angular Measure and Angular Velocity

For a particle moving on a circle of radius r :

- Arc length $s = r\theta$ (with θ in radians).
- Angular velocity $\omega = \dot{\theta}$ (rad/s).
- Linear speed $v = r\omega$.
- Period $T = 2\pi/\omega$, frequency $f = 1/T = \omega/(2\pi)$.

2. Centripetal Acceleration and Force

A particle moving in a circle at constant speed v has acceleration directed towards the centre:

$$a = \frac{v^2}{r} = r\omega^2, \quad v\omega \text{ (equivalent forms)}$$

The force required to produce this acceleration is the **centripetal force**, always directed towards the centre:

$$F = \frac{mv^2}{r} = mr\omega^2.$$

OCR Exam Tip: "Centripetal force" is not a separate force — it's the *net* inward force. For a conical pendulum, that's the horizontal component of tension; for a car on a banked track, the horizontal

component of the normal reaction (plus friction).

3. Horizontal Circular Motion Examples

Object on a Horizontal Turntable

Friction provides the centripetal force. For slipping not to occur: $\mu_s mg \geq mr\omega^2$, giving $\omega_{\max} = \sqrt{\mu_s g / r}$.

Conical Pendulum

A particle on a string of length ℓ moves in a horizontal circle of radius $r = \ell \sin \theta$, where θ is the angle from the vertical. Resolving vertically and horizontally:

$$T \cos \theta = mg, \quad T \sin \theta = mr\omega^2 = m\ell \sin \theta \omega^2.$$

Divide: $\tan \theta = r\omega^2 / g = \ell \omega^2 \sin \theta / g$, i.e. $\cos \theta = g / (\ell \omega^2)$.

Car on a Banked Track

On a track banked at angle α , with no friction the "design speed" is given by

$$\tan \alpha = \frac{v^2}{rg}.$$

4. Vertical Circular Motion

Now speed is *not* constant — there's a tangential component of gravity. Use **conservation of energy** to find v at any point:

$$\frac{1}{2}mv^2 + mgh = \text{constant}.$$

The centripetal equation still holds, using the instantaneous speed:

$$F_{\text{centripetal}} = \frac{mv^2}{r}.$$

Particle on the Inside of a Smooth Vertical Circle

At angle θ from the lowest point, the inward forces are the tension/normal T and a component of gravity:

$$T - mg \cos \theta = \frac{mv^2}{r} \quad (\text{below the horizontal})$$

$$T + mg \cos(\pi - \theta) = \frac{mv^2}{r} \quad (\text{above the horizontal, adjusting signs})$$

At the top: $T + mg = mv_{\text{top}}^2/r$. For the string/normal not to go slack: $v_{\text{top}}^2 \geq gr$.

A particle of mass m is attached to a string of length r and projected horizontally from the lowest point with speed u . Find the minimum u for it to complete a full vertical circle.

Energy at top: $\frac{1}{2}mv^2 + mg(2r) = \frac{1}{2}mu^2$.

Condition at top: $v^2 \geq gr$.

So $u^2 \geq gr + 4gr = 5gr$, i.e. $u \geq \sqrt{5gr}$.

Particle on Outside of a Smooth Sphere

Normal reaction acts outward. At angle θ from the top:

$$mg \cos \theta - R = \frac{mv^2}{r}.$$

Particle leaves the sphere when $R = 0$, i.e. when $v^2 = gr \cos \theta$. Combined with energy conservation, this gives the angle of departure.

5. Three Common Questions

1. Given the speed/angular speed, find tension/normal (set up the radial equation).
2. Given a condition (string slack, particle leaves surface), find the critical speed/angle.
3. Use energy conservation between two positions to find speed at the second.

6. Exam Style Strategy

- Always sketch showing forces acting at the point of interest.
- Choose axes along and perpendicular to the radial direction.
- For vertical motion, use energy for speed, force equation for tension/normal.
- For "just complete" a circle: set $T = 0$ (or $R = 0$) at the top; don't set $v = 0$.